

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE
[A GOVERNMENT AUTONOMOUS COLLEGE]
JGEC/B.TECH/ CE/EE/ME/ ES-CS101/ 2024-25
2024

Programming for Problem Solving

Times: 3 Hours

Full Marks: 70

The figures in the margin indicate full marks.
Candidates are instructed to write the answers in their own words as far as practicable.

GROUP-A
[OBJECTIVE TYPE QUESTIONS]

5x2=10

Answer **all** questions

1. What do you mean by type casting? [2]
2. What do you understand the associativity of operators? [2]
3. What will be the output of following code snippet: [2]
`int i=5;
printf("%d %d %d", i, ++i, i++);`
4. What do you mean by Application Software? Give two examples of Application Software. [2]
5. Convert the hexadecimal number (DCAF.5A)₁₆ into equivalent binary number. [2]

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

4x15 = 60

Answer any **four** questions

6. i) Write down the entry-controlled loop and exit-controlled loop statements with suitable flowcharts and examples. [5]
ii) Explain break and continue statements with suitable examples. [5]
iii) Write a program in C Language to print all prime numbers in a given range. [5]
7. i) Define ternary operators? Write a C program to find largest number among three numbers using ternary operators. [1+4]
ii) What do you mean by storage class? Explain static and automatic storage class with suitable examples. [1+4]
iii) Write a program in C language to arrange all elements in an array in descending order. [5]
8. i) What is recursion? Compare recursive function and iterative function with suitable examples. [1+4]
ii) Write a C program to print the following series: [5]
`0 1 1 2 3 5 8 13` up to N_{th} term. Where the value of N is given as input. [5]
iii) Write a program in C language to display the frequency of a given item in an array. [1+4]
9. i) What do you mean by function? State the advantages of function in C program. [5]
ii) Explain the "call by value" and "call by address" mechanism for passing data from one function to another function with suitable examples. [5]
iii) Write a C program to reverse a given string without using any string library function. [6]
10. i) State the various modes of file operation in C programming. [7]
ii) Write a C program to copy the contents of one file to another file. [7]
iii) Explain the functionality of strcmp() function. [2]

11. i) Explain the differences between structure and union. [3]
ii) Write a program in C language to find the sum of two given matrices and print the resultant matrix. [7]
iii) Write a C program to read N integers into an array A and find the (i) sum of odd numbers, (ii) sum of even numbers [5]

2. i) Explain function call, function definition and function prototype with examples [6]
ii) Write a C program to print the following pattern for 'n' lines. Where the value of 'n' is given as input:

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

[5]

- iii) Write the output of the following code and explain the output [4]

```
int main() {
    char *ptr;
    char str[]="abcdefg";
    ptr=str;
    ptr+=5;
    printf("%s",ptr);
}
```


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JGEC/B.TECH / CE, EE, ME, ECE / BS-M 101B /2024-25
2024
Mathematics - IB

Full Marks: 70

Times: 3 Hours

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GROUP-A
[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions

5 × 2 = 10

1. Find the radius of convergence of the series $\frac{1}{2}x + \frac{1.3}{2.5}x^2 + \frac{1.3.5}{2.5.8}x^3 + \dots$ 2
2. Examine the function $f(x) = x^{\frac{1}{3}}$ for its extreme at $x = 0$. 2
3. Examine the convergence of the improper integral $\int_0^1 \frac{1}{x^2} dx$. 2
4. Show that $\det A =$ product of eigen values of A . 2
5. Find the directional derivative of the surface $x^2 + 2y^2 + 3xz = 0$ at the point $(1,1,1)$ in the direction of the vector $2\hat{i} + 3\hat{j}$. 2

GROUP-B
[LONG ANSWER TYPE QUESTIONS]

Answer any *five* questions

12 × 5 = 60

6. (a) Define a monotone sequence. Show that the sequence $\{u_n\}$ where $u_n = \frac{3n+1}{n+2}$ is monotone increasing. Is the sequence convergent? Justify your answer. 2+1+1
(b) State Raabe's test for convergence or divergence of an infinite series of positive terms and use it to determine whether the infinite series $\sum_{n=1}^{\infty} \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots (2n)} \frac{1}{2n+1}$ is convergent or not. 1+3
7. (a) What is an alternating series? State Leibnitz's test for the convergence of an alternating series and hence show that the series $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$ converges. 1+1+2
7. (a) State Maclaurin's theorem with Lagrange's form of remainder. Expand the function $f(x) = \log_e(1+x)$ in Maclaurin's series, assuming the conditions under which the expansion is valid. 1+3
(b) Obtain the cosine series for $f(x) = x$ in $0 \leq x \leq \pi$ and show that $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$ 2+2
(c) Show that the series $x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ is absolutely convergent when $|x| < 1$ and conditionally convergent when $x = 1$. 2+2
8. (a) State and prove Cauchy's mean-value theorem. 1 1+3
(b) Discuss the applicability of Lagrange's mean-value theorem for the function $f(x) = x(x-1)(x-3)$, $0 \leq x \leq 4$. 2 4
(c) Evaluate $\lim_{x \rightarrow 0} \left[\frac{1}{x^2} - \frac{1}{\sin^2 x} \right]$ 4
9. (a) If $\frac{x^2}{a^2+u} + \frac{y^2}{b^2+u} + \frac{z^2}{c^2+u} = 1$ where u is a function of x, y, z , then prove that $\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2 = 2\left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z}\right)$. 4
(b) If $u = \tan^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{1}{2} \sin 2u$. 4

(c) Find t so that the vector $\vec{f} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x + tz)\hat{k}$ is solenoidal. 4

(a) Prove that $\int_0^1 \frac{dx}{(1-x^6)^{\frac{1}{6}}} = \frac{\pi}{3}$. 4

(b) Prove that $B(m, n) = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta, m > 0, n > 0$. 4

(c) Prove (without expanding):

$$\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ca - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix} = (a^3 + b^3 + c^3 - 3abc)^2.$$
 4

(a) For the matrix $A = \begin{pmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{pmatrix}$, find a matrix P such that $P^{-1}AP$ is a diagonal matrix. 3 6

(b) Determine the condition for which the system of equations

$$x + y + z = 1$$

$$x + 2y - z = b$$

$$5x + 7y + az = b^2$$

admits of (a) only one solution, (b) no solution, (c) infinitely many solutions. 4 6

(a) $A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$. Use Cayley-Hamilton theorem to find A^{100} . 1 4

(b) If A and P be both $n \times n$ matrices and P be non-singular, then prove that A and $P^{-1}AP$ have same eigen values. 4

(c) Reduce the matrix $\begin{pmatrix} 2 & 0 & 4 & 2 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \end{pmatrix}$ to a row reduced echelon form and hence find its rank. 4 4

13. (a) If $f(x, y) = \begin{cases} xy & \text{if } |x| < |y| \\ -xy & \text{if } |x| \geq |y| \end{cases}$, then show that $f_{xy}(0,0) \neq f_{yx}(0,0)$. 4

(b) If $z = f(x, y)$ where $x = e^u \cos v$ and $y = e^u \sin v$, then show that $y \frac{\partial z}{\partial u} + x \frac{\partial z}{\partial v} = e^{2u} \frac{\partial z}{\partial y}$. 4

(c) Examine for extreme value of the function $x^2 + y^2 + (x + y + 1)^2$. 4

30
0, -1, 2
1/1
2

20 22
3 2 6 5
5 2 10 7

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE

[A GOVERNMENT AUTONOMOUS COLLEGE]

COE/B.TECH./CE/EE/ME/BS-PH101/2024-25

2024

PHYSICS

Full Marks: 70

Times: 3 Hours

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GROUP-A

[OBJECTIVE TYPE QUESTIONS]

Answer *all* questions.

5x2=10

1. Does a blackbody always appear black? Explain the term **blackbody**. 2
2. What is the **Poynting vector**? 2
3. Sodium light has two wavelengths which are very closed to each other. In Newton's ring experiment using sodium light, it is found that around 500th ring, maxima and minima corresponding to two components coincide with each other. Calculate the difference between the wavelengths present in sodium light. [$\lambda_{av} = 5893 \text{ \AA}$] 2
4. A force $\vec{F}$ acting on a particle in the xy-plane is given by $\vec{F} = (x^2 + y^2)\hat{i} + cxy\hat{j}$, where c is a constant. For what value of c is the force conservative? 2
5. Define **optic axis** of doubly refracting crystal. 2

GROUP-B

[LONG ANSWER TYPE QUESTIONS]

Answer any *four* questions.

4x15=60

6. i) A particle of mass m moves along the x -axis, attracted towards a fixed point O on it by a force proportional to the distance from O . Initially the particle is at distance x_0 from O and is given a velocity v_0 away from. Determine (a) the position at any time. (b) the velocity at any time. (c) the amplitude, period, frequency and maximum speed. [constant of proportionality is k] 2+2+4
- ii) A body cool down from 60° C to 55° C in 30 s. Using Newton's law of cooling, calculate the time taken by same to cool down from 55° C to 50° C . Assume that the temperature of the surrounding is 45° C . 4
- iii) Position of a particle executing under damped oscillation is given by $x = 20e^{-2t}(\cos 2t + \sin 2t)$. Find the logarithmic decrement. 3
7. i) Find the constants a and b so that the surface $ax^2 - byz = (a + 2)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at the point $(1, -1, 2)$. 3
- ii) Show that $\vec{\nabla} r^n = nr^{n-2}\vec{r}$. 3
- iii) Prove that if $\vec{A}$ and $\vec{B}$ are irrotational, then $\vec{A} \times \vec{B}$ is solenoidal. 3
- iv) Calculate $\int \vec{r} \times d\vec{r}$ along the curve $x = t, y = t^2, z = t^3$ from $t = 0, t = 1$. 3
- v) Show that $\vec{\nabla} \times \vec{\nabla} \phi = 0$ 3

8. i) Obtain the unit vector along the direction of propagation for a wave represented by: $\psi(x, y, z, t) = \text{acos}(2x + 3y + 4z - 5t)$. Where x, y, z are measured in centimeter and t in second. What will be the wavelength and frequency of the wave. 2+1+1
- ii) On the surface of the earth we receive about 1.33 kW of energy per square meter from the sun. Calculate the electric field associated with the sunlight (on the surface of the earth) assuming that it is essentially monochromatic with $\lambda = 6000 \text{ \AA}$. 4
- iii) Using Maxwell's equations, derive electromagnetic wave equation in vacuum and prove that wave propagate with speed of light. 4
- iv) Show that $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ 3
9. i) Discuss the state of polarization when the x and y components of the electric field are given by: $E_x = E_0 \sin(\omega t + kz)$; $E_y = E_0 \cos(\omega t + kz - \frac{\pi}{2})$. 3
- ii) For calcite the values of n_o and n_e for $\lambda = 4046 \text{ \AA}$ are 1.68134 and 1.49694 respectively; corresponding to $\lambda = 7065 \text{ \AA}$ the values are 1.65207 and 1.48359, respectively. We have a calcite quarter wave plate corresponding to $\lambda = 4046 \text{ \AA}$. A left circularly polarized beam of $\lambda = 7065 \text{ \AA}$ is incident on this plate. Obtain the state of polarization of the emergent beam. 4
- iii) Explain the intensity distribution due to Fraunhofer diffraction at a single slit. 5
- iv) In Fraunhofer diffraction due to a single slit, the screen is placed 2 m away from the slit. If the slit width is 0.2 mm and first minima lies 5 mm on either side of the central maxima, find the wavelength of the light. 3
10. i) Show that $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$. 4
- ii) Derive Clausius-Mossotti equation. 4
- iii) Show that a lossy capacitor is represented by a parallel R-C combination. 4
- iv) An atom having polarizability 10^{-40} F.m^2 is at a distance 1 nm from an electron. Calculate the dipole moment induced in the atom. 3
11. i) Consider an x-ray beam with $\lambda = 1.00 \text{ \AA}$ and a γ -ray beam from a ^{137}Cs with $\lambda = 1.88 \times 10^{-2} \text{ \AA}$. If the radiation scattered from free electron is viewed at 90° to the incident beam: (a) What is the Compton wavelength shift in each case? (b) What kinetic energy is given to a recoiling electron in each case. 4
- ii) The mass of an oxygen molecule is $5.4 \times 10^{-26} \text{ kg}$. If this molecule moves with a speed 500 m/s, calculate the de-Broglie wavelength associated with the oxygen molecule. 2
- iii) Calculate $[x^2, p_x]$, $[x, p_x^2]$ 4
- iv) A particle is confined to an infinitely deep potential well. Derive an expression for the energy eigenvalue of the particle. Sketch the eigen-function for the lowest energy state of the particle in this case. 4+1

Useful constants and results

1. Speed of light: $3 \times 10^8 \text{ m/s}$
2. Planck's constant: $6.626 \times 10^{-34} \text{ J.s}$
3. Electron charge magnitude: $1.6 \times 10^{-19} \text{ C}$, Electron rest mass: $9.109 \times 10^{-31} \text{ kg}$
4. Proton rest mass: $1.672 \times 10^{-27} \text{ kg}$
5. Neutron rest mass: $1.675 \times 10^{-27} \text{ kg}$
6. Free space permittivity: $8.8542 \times 10^{-12} \text{ C}^2/\text{N-m}^2$, Free space permeability: $4\pi \times 10^{-7} \text{ N-sec}^2/\text{C}^2$

μ_0

ϵ_0

$B = \frac{\mu}{\mu_0}$